

Multi-Persona Routing Under Uncertainty: Policy Weights and Request Classification

A corrected account of deterministic local profile routing and adaptive state in the SEA path.

Implemented · Reviewed 2026-09-06 · MLAI Research · Agent Safety

The described source behavior is implemented; customer outcomes and deployment readiness are not established.

Practical context

A corrected account of deterministic local profile routing and adaptive state in the SEA path.

Evaluate assistant workflows with explicit behavior, provider and review boundaries.

Implementation and research boundary

Base local completion selects a persona from keyword-derived signals and renders a template. The adaptive completion function can use persisted router weights when invoked through the SEA learning path. The learning loop owns persistence of the updated state.

This note does not claim a learned classifier of emotions, confidence-thresholded response blending, historical task-fit retrieval or a complete automatic rerouting audit trail. Those broader claims in the historical edition are not established by the cited implementation.

Base route: additive keyword weights

The canonical prior is Abbey 0.40, Aviva 0.30 and ABI 0.30. The router splits ASCII whitespace, trims selected trailing punctuation and applies case-insensitive prefix matches against its keyword table. Each match adds one tenth of the keyword's persona score. A positive total is normalized, then the largest weight wins; ties prefer Abbey, then Aviva, then ABI. These values describe a deterministic heuristic, not a trained probability model.

$$w^{(0)} = (0.40, 0.30, 0.30)$$

$$\widetilde{w}_p = w_p^{(0)} + 0.1 \sum_{k \in \text{matches}(I)} s_{k,p}$$

$$w_p = \frac{\widetilde{w}_p}{\sum_j \widetilde{w}_j}$$

Explicit selection and adaptive state

An exact leading persona address can override keyword routing. The parser accepts forms such as “Aviva, be direct” or “ABI: orchestrate”; mentioning a persona later in prose is different. In the adaptive path, the pure modulator updates an exponential moving average and renormalizes it. The default smoothing factor is 0.3. Serialization and validation are explicit; malformed persisted state falls back to defaults rather than becoming an arbitrary routing distribution.

$$e_t = \text{normalize}(0.3w_t + 0.7e_{t-1})$$

What to test and what the result means

Test neutral text, each explicit address, punctuation and whitespace variants, near ties and corrupted saved state. Keep the original utterance separate from retrieved context so evidence text cannot silently impersonate the user's persona selector. The SEA loop owns loading and saving adaptive state; base completion remains store-independent. Compare selected profiles and returned outputs across repeated runs at the pinned revision. A deterministic match establishes routing behavior, not empathy, expertise, calibrated uncertainty or automatic text blending between separate models.

Evaluation before adoption

For a customer pilot, choose a concrete task and identify whether it uses local templates or an explicitly configured provider. Evaluate task correctness, escalation and data handling with representative inputs before making quality claims.

Limitations

Local completion renders persona templates; it is not evidence of a trained foundation model.

Keyword-based checks cannot establish general safety or emotional understanding.

This note does not claim a learned classifier of emotions, confidence-thresholded response blending, historical task-fit retrieval or a complete automatic rerouting audit trail. Those broader claims in the historical edition are not established by the cited implementation.

Pinned implementation sources

Local completion and adaptive routing

<https://github.com/donaldfilimon/abi/blob/6321a47bf4c48a5f58caf2df0eb631b5a2ecdcee/crates/abi-ai/src/completion.rs>

Deterministic persona router

<https://github.com/donaldfilimon/abi/blob/6321a47bf4c48a5f58caf2df0eb631b5a2ecdcee/crates/abi-ai/src/router.rs>

Learning loop and persisted router weights

https://github.com/donaldfilimon/abi/blob/6321a47bf4c48a5f58caf2df0eb631b5a2ecdcee/crates/abi-sea/src/learn_loop.rs

EMA update and state validation

<https://github.com/donaldfilimon/abi/blob/6321a47bf4c48a5f58caf2df0eb631b5a2ecdcee/crates/abi-ai/src/modulator.rs>

Canonical persona priors

<https://github.com/donaldfilimon/abi/blob/6321a47bf4c48a5f58caf2df0eb631b5a2ecdcee/crates/abi-ai/src/identity.rs>

This edition supersedes the June 2026 PDF. The original remains available as historical material; its broader claims must not be treated as current implementation evidence.